

FINAL PROJECT REPORT [FINAL WEEK]

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Group Number: Group 3

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Progress Summary

This week, we focused on improving the **Manual Robot** shooting mechanism. The previous design implemented a catapult-based launcher mechanism which is not working for large balls. Therefore, we changed the mechanism into a **striking mechanism**. This new mechanism allows us to be able to intake a ball and strike it. This new design provided more effectiveness and consistency in launching big balls.

Both **Manual Robot** and **Auto Robot** were now finalized with nothing changed from the auto robot. However, we have optimized the Manual Robot **gripper mechanism**, where we make the gripper arm parallel to the ground so that it can grip the balls more firmly than before. We added an **intake rail** for the ball to be able to slide to the launcher and prepare to be launched.

For the coding part, the **Manual Robot code** is finished, but the Auto Robot code is still not working. When run separated, it was able to run effectively. However, when all code is combined, the robot does not actually run, and debugging is ongoing.

Problems Encountered

- When designing the striking mechanism the mass of the arm is too heavy for the motor to be able to turn. Therefore, we 3D printed the whole arm to be lightweight.
- The auto robot code is still having an issue, the debugging is still ongoing.